CLASS 8 — SCIENCE · FORMULA SHEET

Chapter 1: **Exploring the Investigative World of Science** | Quick revision, one page

Scientific method

- Steps: observation, question, hypothesis, experiment, analysis of data, conclusion.
- Hypothesis: a tentative, testable explanation.
- Conclusions must be supported by data and open to revision.

Variables and fair test

- Independent variable: the quantity the investigator deliberately changes.
- Dependent variable: the quantity measured as the result.
- Controlled variables: all other factors kept constant.
- A fair test changes only ONE variable at a time.

Measurement and units

- SI unit of length: metre (m); mass: kilogram (kg); time: second (s); temperature: kelvin (K), commonly degree Celsius.
- Time period $T = \dfrac{\text{total time}}{\text{number of oscillations}}$.
- Average $= \dfrac{\text{sum of readings}}{\text{number of readings}}$.
- Instruments: scale, stopwatch, thermometer, measuring cylinder, spring balance.

Errors and reliability

- Parallax error: wrong reading because the eye is not directly in line with the mark.
- Repeat readings and take the mean to reduce random error.
- Record data in tables; plot graphs to see how two variables are related.

MASTERMIND